Udacity
Marketing Analytics

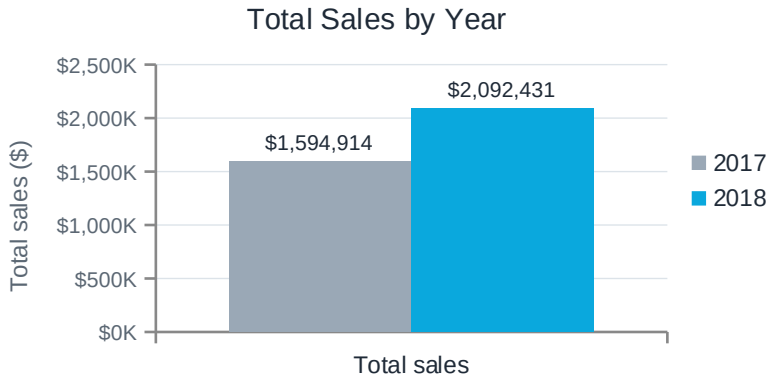
Nanodegree Program
Project: Craft a Report

Black Friday 2017 vs. 2018 Campaign Report

Objective Results

Question: *Did Black Friday 2018 meet the two stated objectives?*

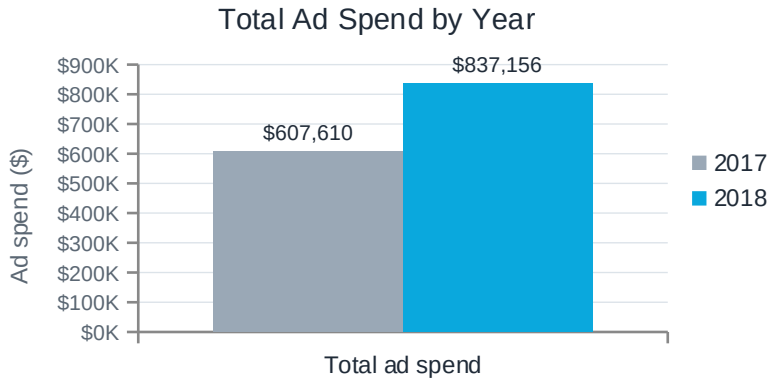
GOAL 1 **Increase total sales by 30% on Black Friday 2018 vs. Black Friday 2017.**



+31.2% **GOAL MET**

Target: +30% (at least $2,073,388). Actual: $2,092,431, which is +31.2% vs. $1,594,914 in 2017.

GOAL 2 **Decrease total ad spend by 30% from Black Friday 2017 to Black Friday 2018.**

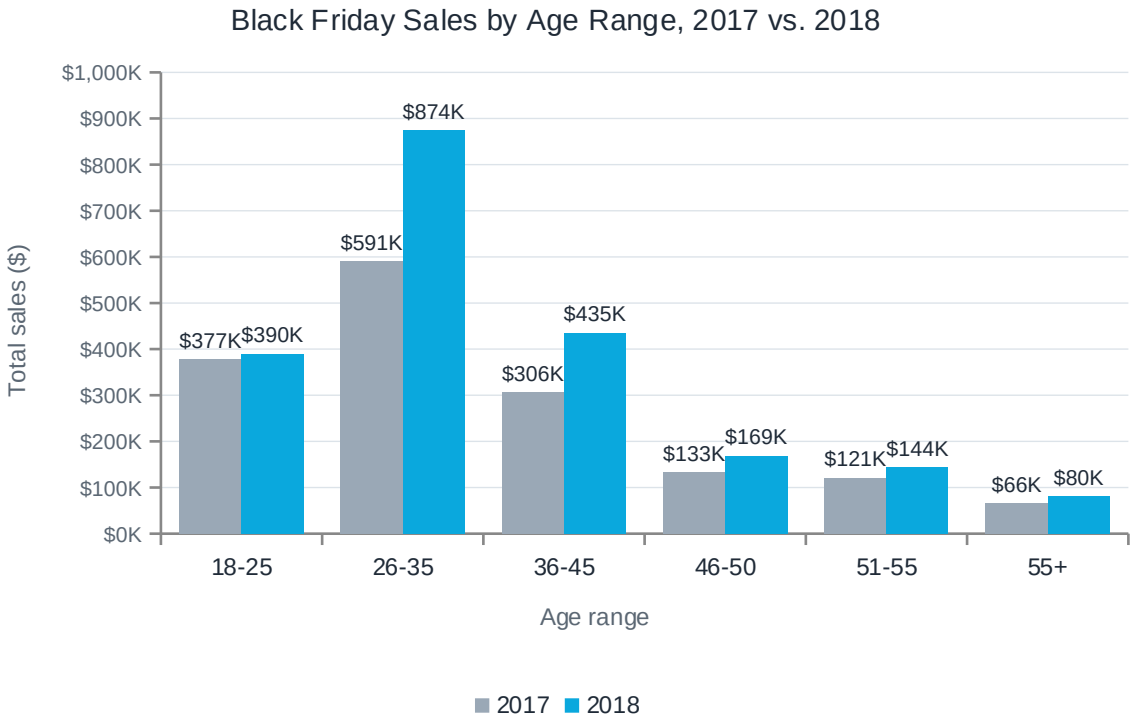


+37.8% **GOAL NOT MET**

Target: -30% (at most $425,327). Actual: $837,156, which is +37.8% vs. $607,610 in 2017: spend went up instead of down.

Evaluate the Audience

Question: *What were total sales for each age range, in 2017 and in 2018?*

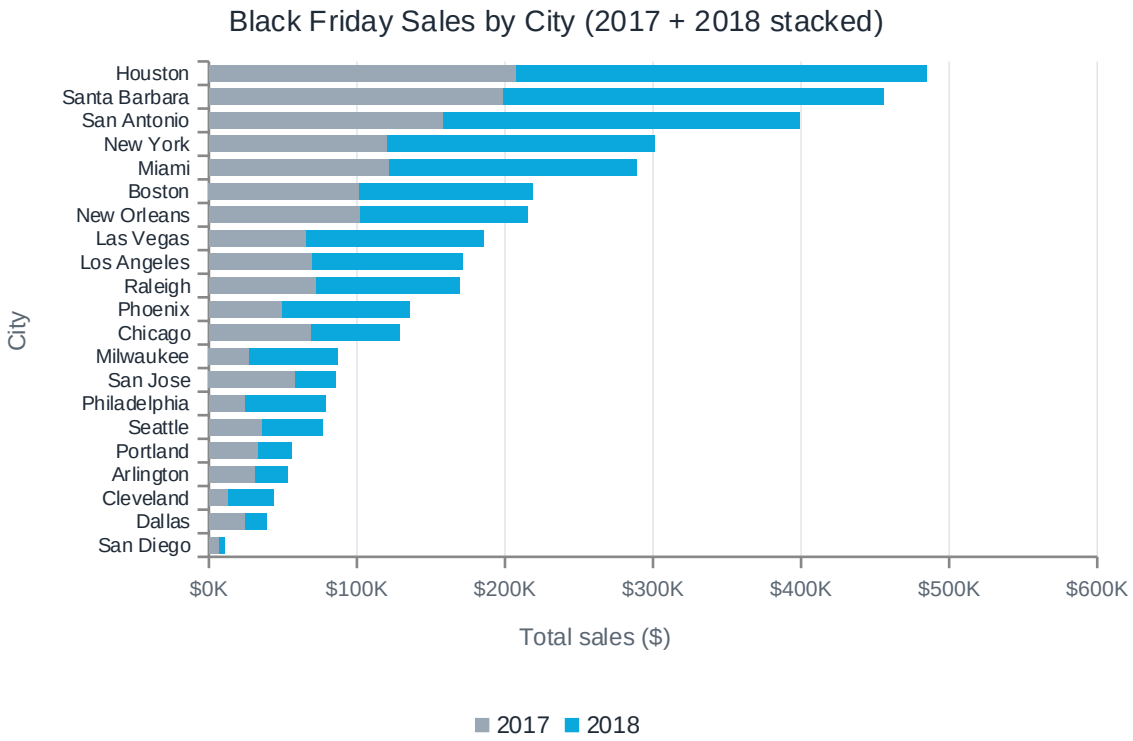


What the data shows

- 26-35 is the top age range in both years: $590.8K in 2017 and $874.2K in 2018 (37% and 42% of all sales).
- Growth was fastest in 26-35 (+48%), then 36-45 (+42%, from $306.5K to $435.2K).
- 18-25 grew least (+3.3%); 55+ is smallest at $80.5K in 2018.
- Why this chart: clustered columns put 2017 and 2018 side by side for each age range, so rank and growth read at once.

Evaluate the Audience

Question: *Which city generated the most sales (and the least)?*

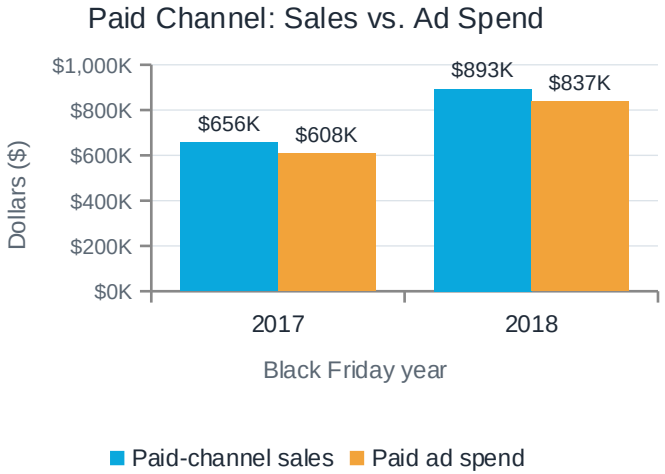


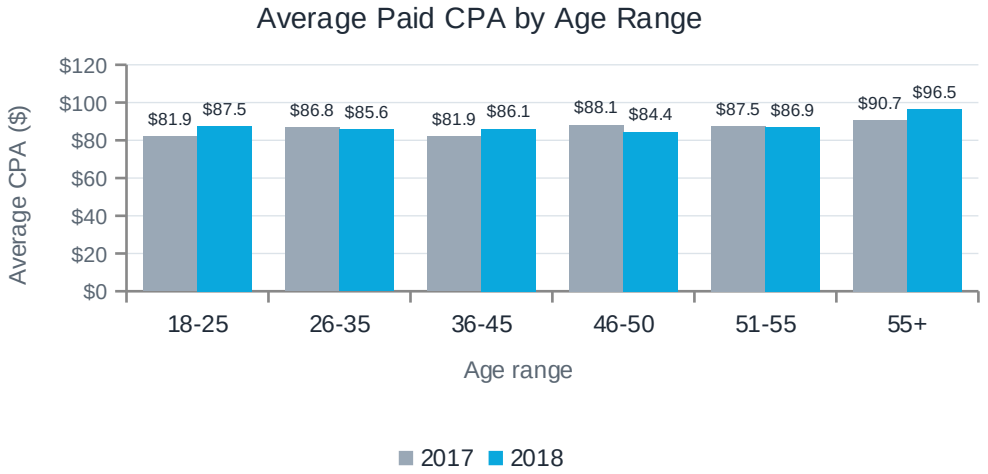
What the data shows

- Most sales: Houston, $485.2K combined ($207.7K in 2017, $277.5K in 2018).
- Santa Barbara ($456.2K) and San Antonio ($398.9K) follow.
- Least sales: San Diego, only $11.0K combined.
- Why this chart: a sorted horizontal bar handles 21 city names legibly; stacking 2017 and 2018 shows both the total ranking and each year's share.

Evaluate the Marketing

Question: *Was the ROI on our Paid channel positive or negative, and which age range had the best CPA?*





PAID ROI, 2017

+8.0%

Positive. Spend $607,610 returned $656,431 in sales.

PAID ROI, 2018

+6.7%

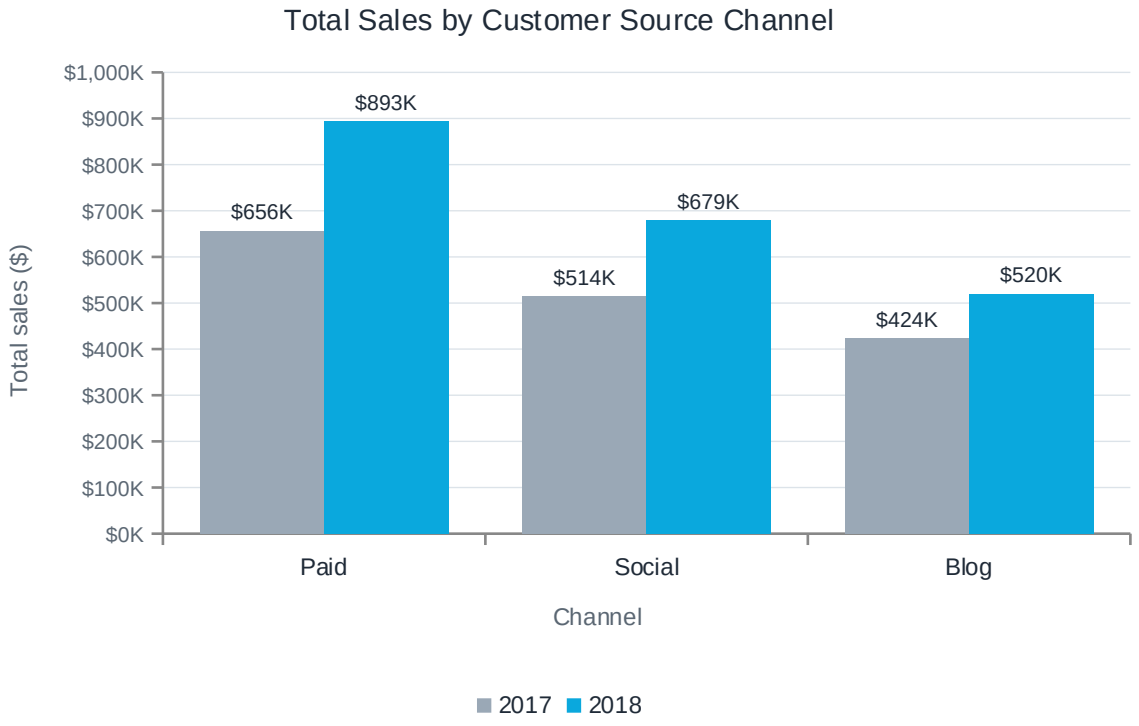
Positive, but lower. Spend $837,156 returned $893,189.

Best CPA (lowest): 18-25 in 2017 ($81.87) and 46-50 in 2018 ($84.42).
Why these charts: columns put sales next to spend so ROI is visible, and clustered columns compare CPA across age ranges and years.

ROI = (paid sales - paid ad spend) / paid ad spend. CPA = ad cost per paid order.

Evaluate the Marketing

Question: *Which channel was the biggest driver of sales in 2017? In 2018?*

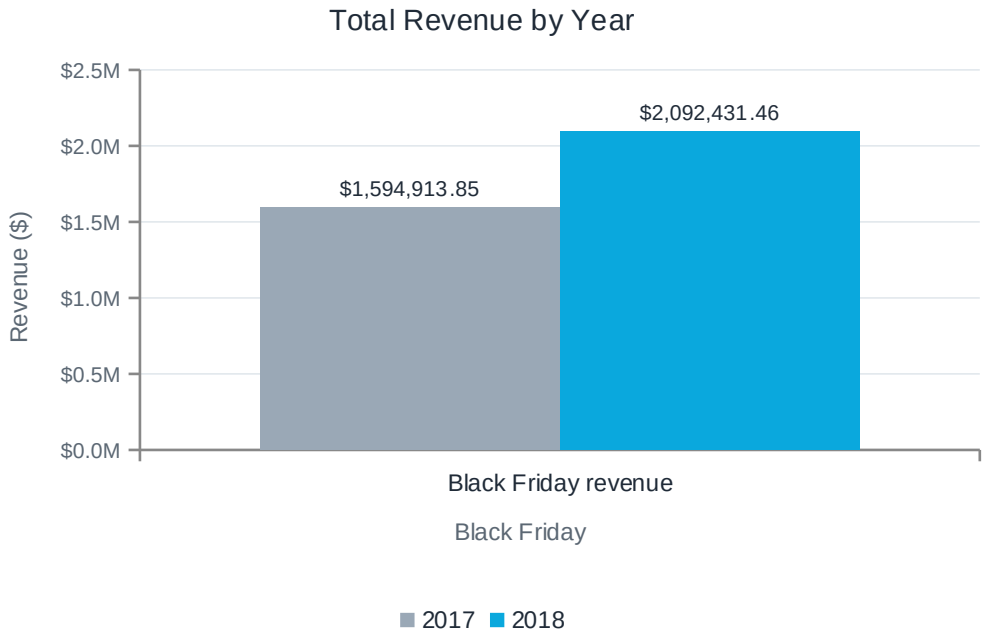


What the data shows

- Paid was the biggest driver in both years: $656.4K in 2017 (41.2% of sales) and $893.2K in 2018 (42.7%).
- Social was second ($514.4K then $679.3K), and Blog third ($424.0K then $520.0K).
- The ranking did not change between years, and all three channels grew.
- Why this chart: clustered columns compare channels and years directly. Paid brings the most sales but is the only channel with ad cost (see ROI slide).

Evaluate the Sales

Question: *How much revenue did we generate in 2017? In 2018?*



Why this chart: two columns on a zero-based axis make the year-over-year jump in revenue easy to compare.

2017 REVENUE

$1,594,913.85

2018 REVENUE

$2,092,431.46

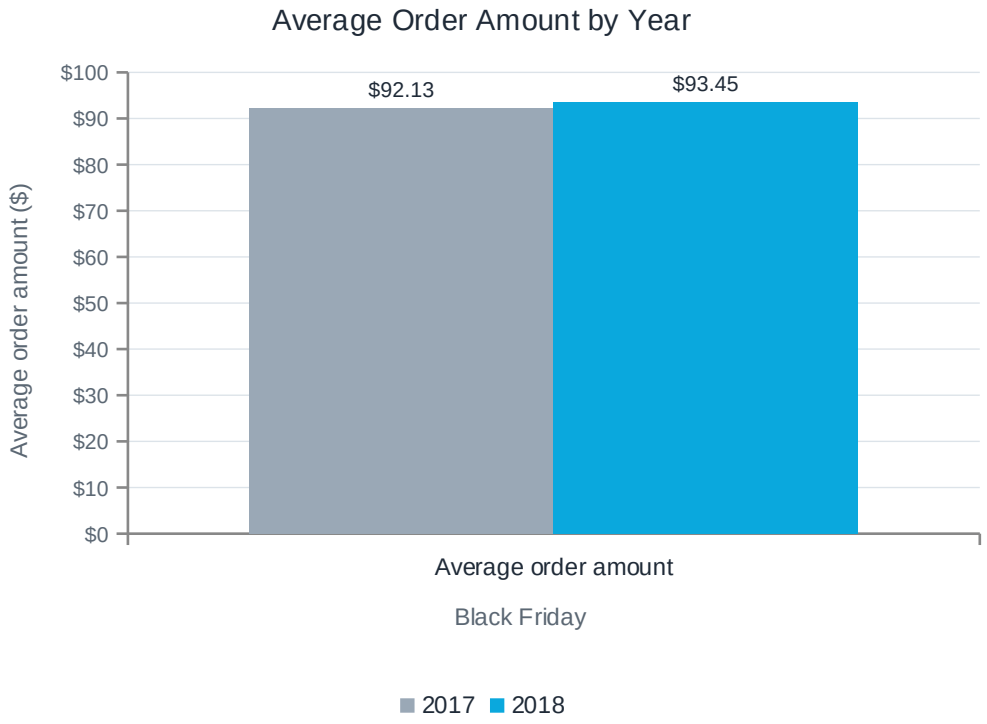
CHANGE, 2017 TO 2018

+31.2%

$497,517.61 more revenue. Revenue is the sum of every order amount in the year.

Evaluate the Sales

Question: *What was our average order amount in 2017 vs. 2018?*

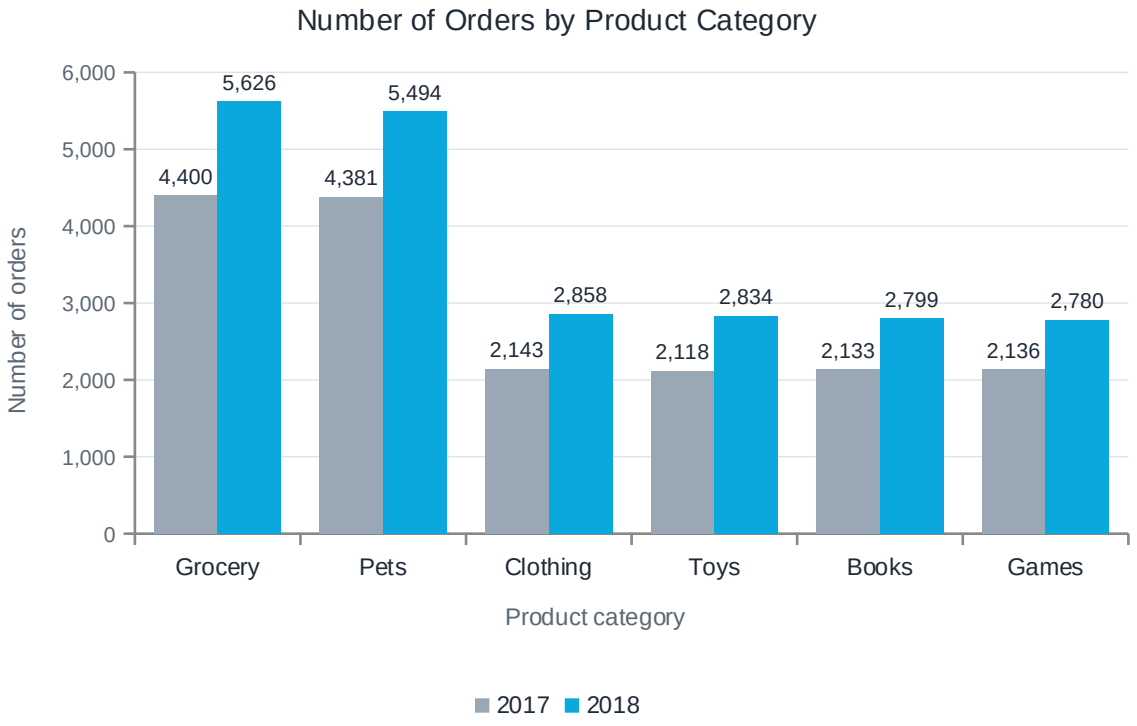


What the data shows

- Average order amount: $92.13 in 2017 and $93.45 in 2018, a change of +1.4%. Basically flat.
- The sales growth came from volume, not bigger baskets: orders rose from 17,311 to 22,391 (+29.3%), and buying customers from 2,304 to 3,031 (+31.6%).
- Why this chart: two columns on a zero-based axis show honestly how small the change is.

Evaluate the Product Categories

Question: *Which product category was most popular in 2017 and in 2018?*

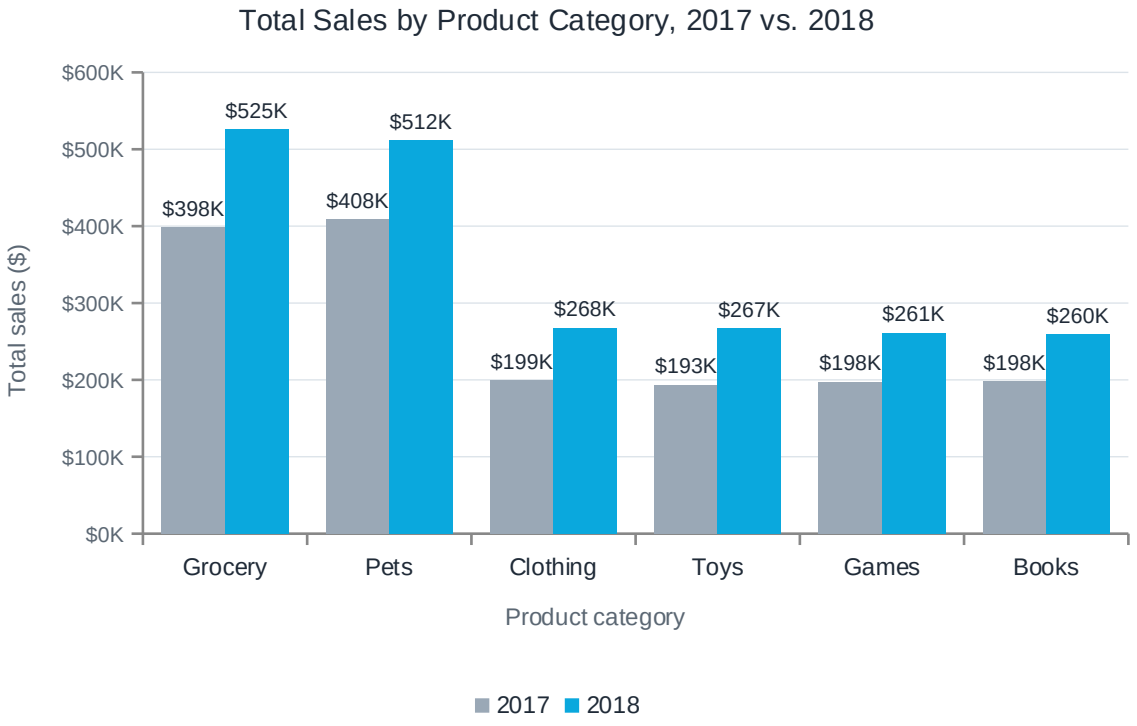


What the data shows

- Most popular (most orders) in both years: Grocery, with 4,400 orders in 2017 and 5,626 in 2018.
- Pets is a very close second (4,381 and 5,494 orders).
- By dollars, Pets edged Grocery in 2017 ($408.5K vs. $398.2K); Grocery led in 2018 ($525.4K vs. $511.7K).
- The other four categories are about half the size of the top two.
- Why this chart: counting orders per category, side by side for both years, shows popularity and its change.

Evaluate the Product Categories

Question: *How did sales break down by product category?*

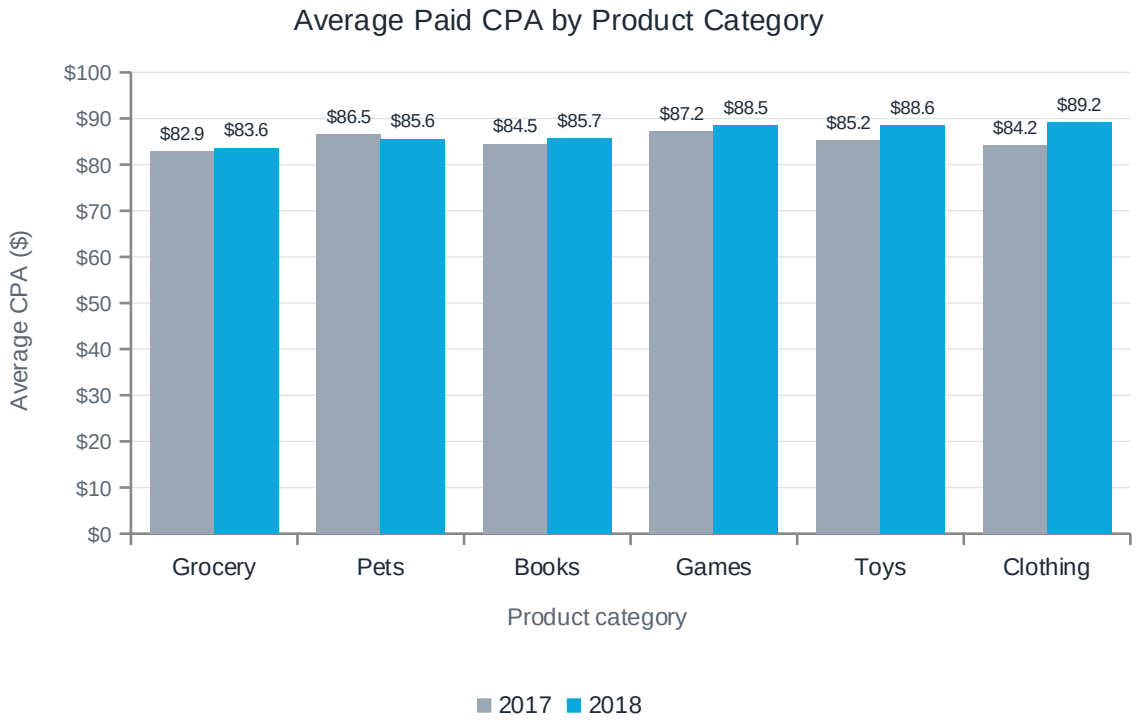


What the data shows

- Grocery and Pets together made up 51% of sales in 2017 and 50% in 2018.
- Every category grew. Fastest: Toys (+38%); slowest: Pets (+25%).
- Why this chart: clustered columns show each category's size and its change at the same time.

Evaluate the Product Categories

Question: *What was the CPA for each product category?*



What the data shows

- Grocery was the cheapest category to acquire a customer in both years: $82.90 (2017) and $83.64 (2018).
- Highest CPA: Games in 2017 ($87.21) and Clothing in 2018 ($89.18).
- CPA rose in five of six categories in 2018, but all categories stay within a few dollars of each other.
- CPA is only recorded for Paid-channel orders, so this covers paid acquisition only.

Everything Else

Backup research, definitions and context

Metric	2017	2018	Change
Total sales	$1,594,913.85	$2,092,431.46	+31.2%
Total ad spend (sum of CPA)	$607,610.41	$837,155.55	+37.8%
Orders (order lines)	17,311	22,391	+29.3%
Unique customers	2,304	3,031	+31.6%
Paid orders	7,150	9,699	+35.7%
Average paid CPA	$84.98	$86.31	+1.6%
Paid ROI	+8.0%	+6.7%	-1.3 pts

Definitions and notes

- Sales = sum of Order Amount. Each data row is one order line; averages use all rows.
- Ad spend = sum of the CPA column. CPA is recorded only on Paid-channel rows.
- Paid ROI = (paid sales - paid ad spend) / paid ad spend.
- Goal 1 holds on dollars (+31.2%). Counted by number of orders, growth was +29.3%, just under the 30% target.
- Data checks: no missing values except CPA on non-Paid rows (expected); 3 (2017) and 5 (2018) fully identical rows were kept as separate order lines.